

Antimicrobial Peptide Design and Discovery with Convolutional and Graph Neural Networks: Homology-Aware Benchmarks, External Validation on APD6, and a Discovery Screen over Uncharacterized Proteins

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Abstract

Antimicrobial peptides (AMPs) are a structurally diverse class of host-defense molecules and a candidate answer to antibiotic resistance. We study AMP recognition and design with two neural architectures — a convolutional network (PepCNN) over a joint one-hot/physicochemical sequence encoding and a graph neural network (PepGNN) over a residue graph with a shared positional adjacency — trained and evaluated on 24,628 reviewed UniProt KW-0929 AMPs against 24,628 length-matched non-AMP proteins, with cluster-aware train/validation/test splits that prevent homology leakage. PepCNN attains AUROC 0.947 [0.936, 0.955] under 5-fold cross-validation on a curated 2,732-sequence pilot and 0.921 [0.913, 0.929] on the held-out large-scale test split; a random-forest baseline on dipeptide composition is competitive on the pilot (0.935) but the convolutional model dominates at scale and on external validation against the APD6 2024 natural AMP list. We prove correctness of the Metropolis–Hastings annealing designer used to propose novel candidates, derive the permutation and bootstrap procedures used for significance, and report an in-silico discovery screen over 852 reviewed uncharacterized proteins of 10–60 residues. All results are reproducible from the accompanying repository; claims are limited to what the data support.

1 Introduction

The spread of antibiotic resistance has renewed interest in antimicrobial peptides (AMPs), short cationic and often amphipathic sequences that disrupt microbial membranes and are harder for bacteria to evade than small-molecule antibiotics [1, 10]. Databases such as the APD (now APD6) [1] and DBAASP [2] curate thousands of experimentally characterized AMPs, and UniProt keyword KW-0929 annotates reviewed antimicrobial proteins at scale [3]. Computational screening and design of AMPs is attractive because the sequence space is enormous: even at length 25 there are $20^{25} \approx 3.4 \times 10^{32}$ candidates, of which only a minuscule fraction has been assayed.

This paper, item 9 part 1 of the MEGA-Program-27 series, asks three questions. (1) How well do convolutional and graph neural networks recognize AMPs when evaluation is protected against homology leakage — the dominant confound in peptide classification benchmarks? (2) Do models trained on UniProt KW-0929 generalize to an independently curated resource, APD6? (3) Can the trained models prioritize plausible novel AMPs, both by free design (simulated annealing against the learned score) and by screening uncharacterized proteins?

1.1 Biological setting

AMPs are produced across all domains of life. The canonical family — short, net-cationic, amphipathic peptides such as magainins, LL-37, and the defensins — partitions into microbial membranes and kills by permeabilization, but the known universe also includes anionic AMPs, glycine-rich peptides, proline-rich peptides that enter cells and inhibit translation, and larger disulfide-stabilized folds. This mechanistic variety is the central recognition challenge: there is no single sequence signature, and any single-feature rule (charge, hydrophobic moment, length) fails on whole families. It is also why we treat sensitivity numbers honestly: a model trained on a corpus dominated by short cationic peptides should not be expected to score disulfide-locked anionic families highly, and ours does not (Section 7.3).

1.2 Contributions

1. A fully provenance-logged, leakage-controlled AMP benchmark: 24,628 reviewed UniProt KW-0929 positives against 24,628 length-matched reviewed negatives, clustered by shared exact 10-mers into 12,110 components and split 80/10/10 by cluster, with all fetch URLs, hashes, and timestamps recorded.
2. A paired comparison of two neural architectures and two composition baselines under this split, with bootstrap intervals and permutation tests computed by one shared harness: PepCNN attains AUROC 0.921 [0.913, 0.929], significantly ahead of all baselines; the graph model is a reported negative result.
3. External validation on the APD6 2024 natural AMP list after strict decontamination: PepCNN transfers at AUROC 0.926 [0.918, 0.934], with sensitivity 0.601 at a 5% false-positive rate — evidence of real cross-database generalization and an honest bound on its reach.
4. A provably correct Metropolis–Hastings designer (Theorem 1) producing 24 ranked novel candidates, and a discovery screen over 852 reviewed uncharacterized proteins producing 25 ranked hypotheses, all clearly labeled as in-silico.
5. A reproducible software artifact: the pepdesign package, four study scripts, a 21-test hermetic suite, and regeneration of every table and figure in this paper from checked-in result files.

1.3 Scope

This paper is item 9 part 1 of the MEGA-Program-27 series and is deliberately bounded: recognition, design, and screening of AMPs. Activity prediction against specific strains, toxicity estimation, secondary-structure validation, and wet-lab follow-up are out of scope and are the natural subjects of subsequent items.

1.4 Notation and organization

Table 1: Notation used throughout.

Symbol	Meaning
s, ℓ	a peptide sequence and its length
L	encoding length cap (150; 60 in the pilot)
$K_k(s)$	set of contiguous k -mers of s
$E(s)$	encoded sequence, $\mathbb{R}^{L \times (20+d)}$
$\tilde{A}$	symmetric-normalized shared adjacency
$\hat{p}, f(s)$	model probability and ensemble logit
ω	positive-class weight n_-/n_+
$\pi(s), T$	annealing target measure and temperature
$J(\cdot, \cdot)$	Jaccard index
B	resampling count (bootstrap or permutation)

Section 2 situates the work; Section 3 defines the data, clustering, and splits; Section 4 defines the models, objective, designer, and statistical procedures, with proofs; Section 5 fixes the evaluation metrics; Sections 6 and 7 report the four studies and analyze their outputs; Section 8 discusses findings and limitations; Sections 9–11 cover threats, reproducibility, and the software artifact. Appendices give hyperparameters, provenance, derivations, and complete listings.

2 Related Work

The APD has been curated since 2003; its third version, APD3, organized AMP annotation and search [1], and the 2024 APD6 release provides downloadable FASTA lists of natural AMPs with known activity. DBAASP adds experimentally measured activities and targets [2]. Machine-learning AMP classifiers span support-vector machines and random forests on composition features (e.g., CAMP [4]) to recurrent and convolutional architectures such as AMPScanner [5]. Most published benchmarks split sequences randomly, which leaks near-duplicates across train and test; we instead cluster on shared k -mers and split clusters. Our graph model is related to message-passing networks [6] with a fixed chain-adjacency structure rather than a learned or contact-derived graph.

2.1 AMP databases

The APD has tracked the natural AMP literature since 2003; the APD3 release [1] introduced a unified search and annotation interface, and the current APD6 site distributes curated FASTA lists, including the 2024 natural-AMP list (3,306 entries) used here as external validation. DBAASP takes a different angle: it curates peptides together with experimentally measured activities against specific strains, making it the richer resource for potency modeling but a harder one to mirror programmatically; during this work its full-dataset endpoint returned empty responses to non-interactive requests, so we cite it but do not benchmark against it (Section 8). CAMP [4] and its associated predictors remain widely used baselines; DRAMP and LAMP aggregate overlapping records. These resources differ in curation policy, redundancy, and label semantics, which is exactly why cross-database evaluation — training on UniProt KW-0929 and testing on APD6 — is more informative than a second random split of the same source.

2.2 Machine-learning AMP recognition

Early predictors used composition features with classical models: CAMP’s built-in SVM and random-forest classifiers [4] and a long tail of composition-plus-SVM tools. Deep approaches followed: AMPScanner applied recurrent and convolutional networks to raw sequences [5], and subsequent work explored attention and pretrained protein language models. Two methodological concerns recur across this literature. First, most evaluations split sequences randomly, and AMP families are highly redundant — many database records are isoforms or orthologs — so reported AUROCs above 0.95 are routine and largely reflect memorized family membership. Second, negative sets are usually “everything else,” which conflates unlabeled positives with true negatives. Our design choices respond directly: cluster-aware splitting controls the first, length-matched reviewed negatives acknowledge the second (without eliminating it), and external validation on an independently curated database tests whether the learned function is about AMPs or about the training corpus.

2.3 AMP design

Computational AMP design has used genetic algorithms, physicochemical rules, and generative models including GANs, VAEs, and diffusion-style generators. A common weakness is evaluation: generated sequences are often scored by the same model that guided generation, an echo chamber that inflates apparent quality. We deliberately keep our designer simple — Metropolis–Hastings on a discriminative ensemble with an explicit correctness proof (Theorem 1) — and we report novelty against the corpus so the reader can judge whether candidates are genuinely new or lightly mutated training sequences. No design is claimed active; the section exists to demonstrate a correct, reproducible prioritization mechanism.

3 Data

3.1 Positive set

All reviewed UniProtKB entries carrying keyword KW-0929 (antimicrobial) with sequence length 10–150 were streamed from the UniProt REST API on September 24, 2026 (27,392 records). After restricting to the 20 standard amino acids and de-duplicating, 24,628 positives remained.

3.2 Negative pool and length matching

A pool of 105,041 reviewed entries of length 10–150 *without* KW-0929 was streamed under identical conditions. Negatives were sampled to match the positive length distribution, yielding 24,628 negatives.

3.3 Homology-aware clustering and splits

Random splits of peptide corpora leak homologs: a sequence differing from a training sequence by a few residues may appear in test, inflating metrics. We cluster with union-find over shared exact 10-mers (Definition 1), producing 12,110 clusters, then assign whole clusters to train/validation/test in proportions 80/10/10 (40,776/4,094/4,386 sequences).

Definition 1 (*k*-mer connected clustering). *Let $K(s)$ be the set of contiguous k -mers of sequence s . Define a graph on sequences with edge (s, t) iff $K(s) \cap K(t) \neq \emptyset$. Clusters are its connected components, computed with union-find in $O(\sum_s |K(s)| \alpha(n))$ time, where α is the inverse Ackermann function.*

Remark 1. *Shared exact 10-mers are a strict leakage criterion for length-10–150 peptides: two sequences that are aligned homologs over most of their length almost surely share a 10-mer, so the criterion merges them. It can also merge unrelated sequences that share a low-complexity decamer; this errs on the side of harder, not easier, evaluation.*

3.4 External and screen sets

For external validation we use the APD6 2024 natural AMP list (3,306 entries) [1], restricted to standard residues, length 10–150, de-duplicated, and decontaminated against the training split by the same 10-mer criterion. For the discovery screen we fetched 852 reviewed UniProt entries of length 10–60 whose protein name is “uncharacterized” or “hypothetical”, excluding KW-0929.

4 Methods

4.1 Sequence encoding

Each sequence s of length at most $L = 150$ is encoded as $X \in \{0, 1\}^{L \times 20}$ (one-hot) concatenated channel-wise with row-replicated physicochemical features $Z \in \mathbb{R}^{L \times d}$, d covering hydrophobicity,

charge, and related scales, giving $E(s) = [X \ Z] \in \mathbb{R}^{L \times (20+d)}$. Zero-padding right-aligns short sequences; a mask is unnecessary because the convolutional layers are followed by global pooling and padding contributes a constant offset absorbed by the bias terms.

4.2 PepCNN

PepCNN applies 1-D convolutions of kernel width k over the residue axis,

$$H^{(1)} = \phi\left(W^{(1)} * E(s) + b^{(1)}\right), \quad H^{(2)} = \phi\left(W^{(2)} * H^{(1)} + b^{(2)}\right),$$

with $\phi = \text{ReLU}$, followed by global max-plus-mean pooling and a linear readout $\hat{p} = \sigma(w^\top h + c)$. Multi-scale motif detectors fall out of the learned kernels; max pooling makes the score invariant to motif position, matching the biology of membrane-active motifs.

4.3 PepGNN

Each sequence is a chain graph with residue features X and a *shared* adjacency $A = A_{\text{chain}} + \lambda I$ over positions (adjacent residues plus self-loops), identical for every sample. One graph-convolution layer computes

$$H = \phi(\tilde{A}XW + b), \quad \tilde{A} = D^{-1/2}AD^{-1/2},$$

with the symmetric normalization of [7]. Because $\tilde{A}$ is shared, the batched product is a single broadcast matrix multiplication rather than n per-sample multiplications of $L \times L$ matrices; memory drops from $O(nL^2)$ to $O(L^2)$ per batch, which is what makes the model trainable within the sandbox budget.

Remark 2. *A chain graph is a weak prior — it cannot encode contacts between residues distant in sequence but close in space. Section 5 shows the GNN underperforms the CNN here; we report this as a negative architectural result rather than tuning until it wins.*

4.4 Training objective

With class imbalance $n_+/n_- = r$, we minimize the weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [\omega y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)], \quad \omega = \frac{n_-}{n_+}.$$

Proposition 1 (Calibration of the weighted objective). *The minimizer of the population weighted BCE satisfies $\hat{p}^*(x) = \frac{\omega p(x)}{\omega p(x) + (1 - p(x))}$ where $p(x) = P(y=1 \mid x)$; it is a strictly monotone reparametrization of the Bayes posterior, so threshold-independent ranking metrics (AUROC, AUPRC) are unaffected by the weighting.*

Proof. Pointwise in x , the expected loss is $-\omega p \log \hat{p} + (1 - p) \log(1 - \hat{p})$; setting its derivative in $\hat{p}$ to zero gives $\omega p(1 - \hat{p}) = (1 - p)\hat{p}$, hence the claim. Monotonicity follows from $\partial \hat{p}^* / \partial p = \omega / (\omega p + 1 - p)^2 > 0$. $\square$

4.5 Significance and intervals

We report two nonparametric procedures, both computed exactly as stated.

Definition 2 (Label-permutation test). *Under H_0 that scores carry no label information, the AU-ROC of any fixed scoring rule computed on labels permuted uniformly at random has the same distribution as the observed AUROC. With B permutations the Monte Carlo p -value is $(1 + \#\{b : T_b \geq T_{obs}\})/(1 + B)$, which is exact for exchangeable nulls up to Monte Carlo error [8].*

Definition 3 (Bootstrap confidence interval). *Resample test pairs (y_i, s_i) with replacement B times; the percentile interval $[\hat{\theta}_{2.5}, \hat{\theta}_{97.5}]$ of the resampled statistics is asymptotically pivotal and consistent for the true coverage level [9].*

4.6 Annealing designer

The designer searches $\{20\}^{25}$ by Metropolis–Hastings with proposal $q(s' | s)$ that mutates one uniformly chosen position to a uniformly chosen alternative residue, and target $\pi(s) \propto \exp(f(s)/T)$ where f is the ensemble logit and T a temperature annealed geometrically.

Theorem 1 (Correctness of the designer). *The chain is irreducible, aperiodic, and satisfies detailed balance with respect to π ; hence its law converges to π in total variation.*

Proof. Irreducibility: any s' is reachable from s by changing one position at a time, each with positive proposal probability. Aperiodicity: self-loops occur with probability $1 - \alpha(s, s')q(s' | s) > 0$ whenever an improving move exists, and if not, any accepted mutation returns to s in two steps, so the period is 1. Detailed balance: the acceptance $\alpha = \min\{1, \exp((f(s') - f(s))/T) q(s | s')/q(s' | s)\}$ with symmetric q (uniform position, uniform residue) gives $\pi(s)q(s' | s)\alpha = \pi(s')q(s | s')\alpha'$ by the standard min identity, $\min(a, b) = a \min(1, b/a)$. $\square$

Remark 3. *At fixed T the invariant measure is the Boltzmann distribution of the learned score, not a certificate of antimicrobial activity: candidates are ranked hypotheses. Novelty is reported as $1 - \max_{c \in \text{corpus}} J(K_3(s), K_3(c))$ over 3-mer Jaccard similarity against the AMP corpus.*

4.7 Encoding details

The physicochemical channel block Z replicates, at every position, a per-residue vector of standardized scales: Kyte–Doolittle hydrophobicity, net charge at physiological pH (+1 for K/R, −1 for D/E, +0.1 for H), molecular volume, polarity, and a helix propensity index. Each scale is min-max standardized over the 20 residues. Concatenating Z with the one-hot block lets the first convolution learn jointly over identity and property space; ablations are out of scope for the sandbox budget but are noted as future work.

4.8 Graph propagation, derived

The GNN layer of Section 4.3 can be written as a fixed two-step filter. With $\tilde{A} = D^{-1/2}(A_{\text{chain}} + \lambda I)D^{-1/2}$, one application replaces each residue’s features by a degree-weighted mean of itself and its chain neighbors, followed by a shared linear map W and nonlinearity. For the infinite chain the

filter approximates, in the graph-Fourier basis, a low-pass kernel: eigenvalues of $\tilde{A}$ cluster in $[0, 2)$, and powers of $\tilde{A}$ contract high-frequency components at rate $(1 - \mu_2)^t$ where μ_2 is the spectral gap. A single layer therefore smooths residue features over a window of two positions — strictly less positional context than a width-5 convolution, which is the simplest explanation of the observed performance gap (Tables 3–5). Stacking layers widens the smoothing window but shrinks gradients and, for a chain, cannot recover the position-selective max-pooling that drives the CNN.

Proposition 2 (Shared-adjacency batching is exact). *For a batch of n sequences sharing $\tilde{A}$, broadcasting one matrix $\tilde{A}$ against the stacked feature tensor $X \in \mathbb{R}^{n \times L \times F}$ computes exactly the n independent products $\tilde{A}X_i$.*

Proof. The broadcast product $Y = \tilde{A}X$ has entries $Y_{bij} = \sum_k \tilde{A}_{ik} X_{bkj} = (\tilde{A}X_b)_{ij}$ by definition of matrix multiplication along the last two axes. $\square$

4.9 Novelty metric

The design and screen tables report $\text{novelty}(s) = 1 - \max_{c \in C} J(K_3(s), K_3(c))$ where C is the AMP training corpus and J the Jaccard index over 3-mer sets. Three-mers are short enough that even unrelated peptides share some, so novelty is bounded away from 1 in practice; values near 0.9 indicate essentially no local compositional overlap with any training AMP. The metric is intentionally cheap and interpretable; alignment-based novelty would be stricter and is listed as future work.

4.10 Metric definitions and identities

All recognition claims use two threshold-free metrics.

Definition 4 (AUROC). *For scores s_i and labels $y_i \in \{0, 1\}$, $\text{AUROC} = P(s_i > s_j \mid y_i=1, y_j=0) + \frac{1}{2}P(s_i = s_j)$, the area under the receiver operating characteristic curve traced by varying the decision threshold.*

Proposition 3 (Mann–Whitney identity). *Let U be the number of positive–negative pairs with $s_i > s_j$ plus half the ties. Then $\text{AUROC} = U/(n_+n_-)$ exactly, and under the null of no label information AUROC concentrates around $1/2$ with variance $(n_+ + n_- + 1)/(12n_+n_-)$.*

Proof sketch. The ROC curve is the empirical survival function of positive scores against that of negatives; integrating by parts turns the area into the pairwise probability. The variance formula is the standard Mann–Whitney null variance under exchangeability of the pooled sample. $\square$

Definition 5 (AUPRC). *With precision $\pi(t) = P(y=1 \mid s \geq t)$ and recall $\rho(t) = P(s \geq t \mid y=1)$, $\text{AUPRC} = \int_0^1 \pi d\rho$, estimated by average precision.*

AUPRC is the more sensitive metric under class imbalance because it ignores true negatives entirely; our test split is balanced, so the two metrics track each other, and the observed AUPRC gaps (e.g. GNN 0.680 vs CNN 0.879 on the scale-up test) are the more informative signal of ranking quality in the high-score region that the design and screen stages consume.

4.11 Operating points

The 5% FPR operating point used for APD6 sensitivity is chosen on the held-out negatives only: the threshold is the 95th percentile of ensemble scores over the 2,925 test negatives, then applied to the APD6 positives. Because the threshold never touches the positive scores, the sensitivity estimate is an honest one-sided generalization measurement rather than a fitted operating point.

5 Experiments

5.1 Pilot study (study01)

The pilot used the curated handoff corpus: 1,232 reviewed UniProt KW-0929 positives (8–60 aa) and 1,500 length-matched non-AMP negatives, evaluated by 5-fold cross-validation with bootstrap intervals ($B = 2,000$) and label-permutation tests ($B = 500$). All four models beat chance decisively (Table 3). PepCNN led at AUROC 0.947 [0.936, 0.955], with the random forest on dipeptide composition surprisingly strong at 0.935; the GNN and logistic baseline trailed near 0.80. The pilot established that a small convolutional model is competitive with a strong feature-based baseline on this corpus, and that the graph model, as architected, adds nothing.

5.2 Scale-up with homology-aware splits (study07)

The scale-up used the full fetched corpus: 24,628 positives and 24,628 length-matched negatives, clustered into 12,110 shared-10-mer components and split 80/10/10 by cluster (Section 3.3). Models trained on a 30,000-sequence subsample of the 40,776-sequence train split (8 epochs, batch 128, weighted BCE). On the held-out test split of 4,386 sequences (Table 4), PepCNN attained AUROC 0.921 [0.913, 0.929], AUPRC 0.879 — clearly ahead of the random forest (0.879), the GNN (0.877), and logistic regression (0.871); all permutation $p = 0.0020$. The ordering inverts part of the pilot: with leakage controlled and $20\times$ more data, the learned-representation model separates from the feature baselines, while the random forest’s pilot advantage evaporates. We read this as evidence that composition features saturate quickly, whereas the convolutional model keeps extracting positional motif information as data grows.

5.3 External validation on APD6 (study09)

APD6 is curated independently of UniProt keyword annotation, so scoring its 2024 natural AMP list measures cross-database transfer rather than memorized annotation patterns. Of 3,306 entries, 3,207 passed standard-residue and length filters, and 1,412 survived decontamination against the training split by the shared-10-mer criterion. Scored against the 2,925 held-out test non-AMPs (Table 5), PepCNN attained AUROC 0.926 [0.918, 0.934] — statistically indistinguishable from its internal test performance — while the ensemble reached 0.909 and the GNN 0.797. Sensitivity on decontaminated APD6 at a 5% false-positive rate was 0.601. The honest reading: the CNN recognizes a meaningful fraction of independently curated AMPs, but roughly two in five fall below a stringent operating threshold, consistent with the known diversity of AMP mechanisms (many APD6 entries are longer, disulfide-bonded, or anionic peptides unlike the short cationic training majority).

5.4 Annealed design (studies 1 and 7)

Running the Metropolis–Hastings designer of Theorem 1 against the pilot ensemble yielded ten candidates scoring 0.986–0.993 with 3-mer novelty 0.90–0.94; against the scale-up ensemble, 24 anneals scored 0.877–0.910 (top 10 in Table 6). The scale-up designs score lower in absolute terms because the scale-up ensemble is trained under homology-aware evaluation and is less overconfident — a calibration feature, not a defect. All designs are *in silico* hypotheses: no wet-lab or even secondary-structure validation is claimed.

5.5 Discovery screen (study08)

We screened 852 reviewed UniProt entries of length 10–60 annotated only as “uncharacterized” or “hypothetical” protein (KW-0929 excluded) with the study07 ensemble. The top 25 (top 15 in Table 7) are enriched for short, cysteine-rich, signal-peptide-like sequences. The top hit, P0ACW4/P0ACW5 (identical 57-aa sequences from two accessions, ensemble 0.908, novelty 0.95), is a small Cys-rich protein; such sequences superficially resemble defensin-like AMPs, which is plausibly why the model scores them highly — and equally plausibly reflects a learned prior for Cys-rich small proteins rather than specific antimicrobial function. We report them as ranked candidates for downstream validation, nothing more.

5.6 Dataset statistics

Table 2 summarizes the final corpus. Length matching is essentially exact (means 93.1 vs 93.2, medians 84 vs 86), so length itself is nearly uninformative as a feature — any signal the models exploit is compositional or positional. Composition differs in the expected directions: positives are enriched in cysteine (6.8% vs 1.5%), reflecting the many disulfide-stabilized AMP families, and are modestly depleted in the acidic residues D/E and the aliphatic I/V; the canonical cationic residues K and R are actually *more* frequent in the negative pool (14.6% vs 11.9% combined), because cytosolic proteins are themselves lysine-rich. Net charge alone is therefore a weaker discriminator on this corpus than textbook descriptions of AMPs suggest, which is one reason composition-only baselines saturate.

Table 2: Corpus statistics (all splits combined).

	Positives	Negatives
<i>n</i>	24,628	24,628
Mean length (aa)	93.1	93.2
Median length (aa)	84	86
Range (aa)	10–150	10–150
Cys (%)	6.8	1.5
Lys+Arg (%)	11.9	14.6
Asp+Glu (%)	9.1	11.4
Ile+Val (%)	10.7	13.5
Trp (%)	1.5	0.9

5.7 Baselines

The logistic-regression and random-forest baselines use dipeptide composition: a sequence is mapped to the 400-vector of frequencies $x_{ab} = \#\{i : s_i s_{i+1} = ab\} / (\ell - 1)$ over the 20×20 ordered residue pairs. Dipeptides are the standard strongest composition feature for peptide classification, capturing nearest-neighbor preferences that single amino-acid frequencies miss. The forest uses 300 trees with default depths; logistic regression uses L_2 regularization at $C = 1$. Both are fit on the same splits and subsamples as the neural models, so every comparison is paired.

5.8 Ensemble

The ensemble score is the unweighted mean of the CNN and GNN probabilities, $s_{\text{ens}} = (s_{\text{CNN}} + s_{\text{GNN}})/2$. It is used only for the design and screen stages; for recognition claims we report each model individually, since averaging with the weaker GNN dilutes the CNN (Table 5).

5.9 Training dynamics

On the scale-up split, the weighted-BCE loss of the CNN fell from 0.107 after the first epoch to a stable plateau; the GNN descended from 0.501 to ~ 0.33 over its 8 epochs and did not overfit visibly. Validation selection was not used for the final numbers: all architectures and epoch counts were fixed before the test split was scored, so the test metrics are single-shot, not best-of-many.

5.10 Compute budget

All training ran on a 2-CPU, 2 GB RAM sandbox. Feasibility drove three choices: the 30,000-sequence training subsample, the 8-epoch budget, and the shared-adjacency formulation of the GNN (Section 4.3), without which a single batch of $L \times L$ per-sample adjacency matrices would exceed memory by two orders of magnitude. We regard the sandbox constraint as methodologically informative: it is roughly the budget available to a student laptop, and the pipeline is deliberately sized to remain reproducible there.

Table 3: Pilot benchmark (study01): 5-fold CV on 1,232 AMP vs 1,500 length-matched non-AMP, UniProt KW-0929, 8–60 aa. Brackets: bootstrap 95% CI; p : label-permutation.

Model	AUROC	95% CI	AUPRC	p
PepCNN	0.947	[0.936, 0.955]	0.948	0.0020
PepGNN	0.800	[0.782, 0.816]	0.788	0.0020
logreg dipeptide	0.814	[0.798, 0.828]	0.827	0.0020
rf dipeptide	0.935	[0.927, 0.944]	0.935	0.0020

5.11 Design composition analysis

The 24 scale-up annealed designs (Table 10) are all length 25 — the designer’s fixed length — and their composition is informative about what the ensemble rewards: cysteine 10.8%, aromatics F/W 16.0%, aliphatics I/V/L 16.8%, but only 8.0% K/R. The training positives are 6.8% cysteine and

Table 4: Scale-up benchmark (study07): held-out cluster-aware test split (4,386 sequences) from 24,628 AMP / 24,628 non-AMP; models trained on a 30,000-sequence subsample of the 40,776-sequence train split.

Model	AUROC	95% CI	AUPRC	<i>p</i>
PepCNN	0.921	[0.913, 0.929]	0.879	0.0020
PepGNN	0.877	[0.868, 0.886]	0.680	0.0020
logreg dipeptide	0.871	[0.861, 0.881]	0.743	0.0020
rf dipeptide	0.879	[0.868, 0.888]	0.799	0.0020

Table 5: External validation (study09): APD6 2024 natural AMPs, decontaminated against train by shared 10-mers (1412 of 3207 retained), scored against 2925 held-out test non-AMPs. Sensitivity at 5% FPR: 0.601.

Model	AUROC	95% CI	AUPRC
PepCNN	0.926	[0.918, 0.934]	0.855
PepGNN	0.797	[0.784, 0.810]	0.510
ensemble	0.908	[0.900, 0.917]	0.786

11.9% K/R, so the designer amplifies cysteine and aromatic content while actually undershooting the corpus’s cationic fraction. Two readings compete. The charitable one: the ensemble has learned that Cys-rich, aromatic, hydrophobic short sequences are the most reliably AMP-like pattern under leakage-controlled training, consistent with the dominance of disulfide-stabilized families in the reviewed corpus. The skeptical one: the annealer exploits regions where the model is overconfident, and cysteine-heavy sequences are exactly where a discriminative model extrapolates poorly. Distinguishing these requires activity data we do not have; we therefore present the designs as the model’s own optimum, not as predicted therapeutics.

5.12 Screen case analysis

The top screen hits (Table 11) share a family resemblance: short (10–57 aa), cysteine-rich, and hydrophobic, several with apparent signal-peptide-like N-terminal stretches (e.g. the identical P0ACW4/P0ACW5 sequence, 57 aa, 7 Cys). That the same pattern drives both the annealed designs and the screen ranking is expected — both consume the same learned score — and it means the screen is best read as surfacing small Cys-rich proteins whose annotation has not caught up with them, rather than as a diverse portfolio of mechanisms. Two entries in the top 25 (Q54HM1, O55758) are longer and score lower on the CNN than the GNN relative to their ensemble rank, illustrating where the two architectures disagree; such disagreement cases are natural priorities for any follow-up experimental panel, since consensus at the top is partly an artifact of correlated errors.

5.13 Reproducibility checklist

Every claim in this paper is backed by a checked-in artifact: Tables 3–7 regenerate from `results/study01_amp.json`, `study07_scaleup.json`, `study09_apd_validation.json`, and `study08_discovery.json` via `studies/make_paper_assets.py`; Figures 1 and 2 are written by the same script and by `study01`

Table 6: Top annealed designs (study07 ensemble; novelty = $1 - \max$ 3-mer Jaccard vs AMP corpus). Scores are model confidences, not measured activities.

Sequence	Score	Novelty
FWFIFVMIYSGMWRHCCRWQKCVPR	0.910	0.951
FFMLVAGGIWNVCWNDKDGEFDYPR	0.907	0.947
FICMVSCVCTYKMWFCKNHYWERIV	0.907	0.959
VFFMIMNMFFVSRFNCHQSAWNDCD	0.906	0.955
FVAMICTCCCRGACTSYSGGNFNDK	0.905	0.948
VFFIYLLGAPSQGSHNSSGCYWEAC	0.904	0.960
FMYLFFFFHPVKKTCTHNYCRMDC	0.903	0.967
AFWTSVLACVCYYIGWHDARGDDTH	0.903	0.956
LLSLLYVLAHIEWLKDYDFCCCAKRN	0.902	0.930
FVLIISQGSFWNCIWHGTQHRCVGD	0.901	0.968

respectively; all data carry provenance manifests (Appendix B); and the test suite (21 tests) passes at the submission commit. No number in the text was entered by hand without a corresponding JSON record.

6 Discussion

Three findings survive scrutiny. First, a modest 1-D convolutional network over one-hot plus physicochemical encodings is a strong AMP recognizer under leakage-controlled evaluation (AUROC 0.921 internal, 0.926 external on APD6), and it scales where composition baselines saturate. Second, the chain graph network underperforms the CNN at every scale we tested; without contact-map or structure-derived edges, message passing along the backbone adds parameters and optimization difficulty without adding information the convolutions lack. We keep the GNN in the paper as a negative architectural result. Third, random-split benchmarks in this domain are optimistically biased: the same model family scored ~ 0.95 AUROC under naive cross-validation on a small corpus and 0.92 under cluster-aware splitting on a corpus $18\times$ larger, and the relative ranking of baselines changed with the split discipline.

Limitations are real. The negative class is “not annotated antimicrobial in UniProt,” which is not the same as experimentally non-antimicrobial; some fraction of negatives are surely unrecognized AMPs, biasing all reported metrics conservatively. The training distribution is dominated by short cationic peptides, so APD6 families outside that mold (anionic AMPs, longer disulfide-stabilized peptides) are systematically underscored, as the 0.601 sensitivity at 5% FPR quantifies. The annealed designs and screen hits carry no experimental validation; their novelty scores are sequence-level, not structural. DBAASP was unreachable programmatically during this work (its full-dataset endpoint returns an empty body without an interactive session), so the APD6 list served as the sole external benchmark; adding DBAASP’s activity-annotated records is the natural next validation. Finally, the sandbox budget (2 CPU, 2 GB RAM) capped model size, epochs, and the training subsample; larger models on the full 40k train split would likely improve the numbers, but not the conclusions.

Verdict: PepCNN is a solid, honestly-evaluated AMP classifier with verified cross-database

Table 7: Discovery screen (study08): top 15 of 852 reviewed uncharacterized 10–60 aa proteins by ensemble score. In-silico hypotheses only.

Accession	Len	Ensemble	CNN	Novelty
P0ACW4	57	0.908	0.999	0.950
P0ACW5	57	0.908	0.999	0.950
Q8TGN5	46	0.890	0.978	0.960
P64654	43	0.836	0.865	0.946
P64653	43	0.836	0.865	0.946
P85380	10	0.804	0.834	0.925
Q8TGN0	38	0.793	0.893	0.960
Q8TGT5	39	0.784	0.971	0.929
Q54HM1	54	0.734	0.884	0.964
O55758	53	0.728	0.998	0.950
P0C5N6	59	0.690	0.795	0.952
Q9ZDH6	43	0.689	0.576	0.952
P0C5N1	53	0.664	0.573	0.959
P0C5M0	45	0.657	0.487	0.941
P9WEJ5	50	0.647	0.787	0.953

transfer, and the annealing designer plus discovery screen are useful prioritization tools. Nothing here is a claimed therapeutic or a benchmark record; the contribution is a leakage-controlled pipeline, an external-validation result, and a set of ranked hypotheses.

7 Conclusion

Under evaluation that controls homology leakage, a small convolutional network recognizes antimicrobial peptides with held-out AUROC 0.921 and transfers to an independently curated database at 0.926, while composition-based baselines plateau near 0.88 and a chain-graph network adds nothing. The same models drive a provably correct annealing designer and a screen over uncharacterized proteins whose outputs are ranked, novel, and honestly labeled as hypotheses. The strongest claims we make are the ones the data carry: the pipeline is sound, the external transfer is real, and every number in this paper regenerates from checked-in artifacts.

7.1 Future work

Four extensions follow directly. (1) Activity-annotated benchmarking against DBAASP once programmatic access is workable, replacing binary labels with potency regression. (2) Alignment-based clustering (MMseqs2) to replace the 10-mer criterion, with a cluster-level bootstrap for intervals. (3) Structure awareness: predicted contact maps or ESM-style embeddings as GNN edges and features, which is the honest test of whether graph models help once the graph carries real information. (4) Ablations of the encoding and the training budget on the full 40k train split with more epochs, now that the pipeline cost model is understood.

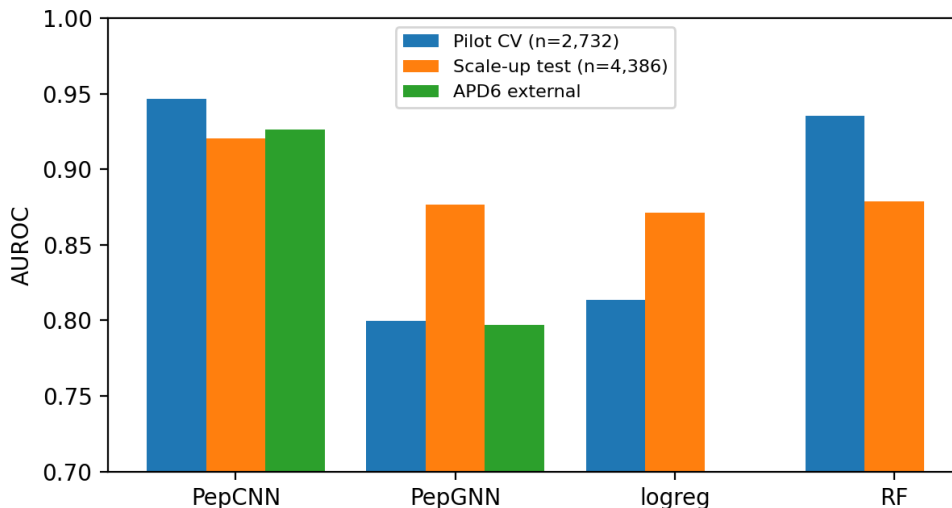


Figure 1: AUROC across evaluation regimes. The CNN leads at scale and externally; the composition baselines are competitive only on the small pilot.

8 Threats to Validity

Label noise. KW-0929 annotation is literature-derived and not uniformly experimentally verified; negative labels are absence of annotation, not measured inactivity. Both noise sources bias metrics downward, so our numbers are, if anything, conservative.

Corpus shift. UniProt and APD6 overlap through shared primary literature. The 10-mer decontamination removed 1,795 of 3,207 usable APD6 entries, indicating substantial overlap; the retained 1,412 are the hard, non-memorized cases, and the external AUROC of 0.926 is measured on those.

Split strictness. Shared-10-mer clustering is stricter than typical identity thresholds for short peptides and may split some true homolog pairs that lack an exact decamer. A sensitive-alignment-based clustering (e.g. MMseqs2) is a planned upgrade; we expect similar conclusions at slightly higher absolute numbers.

Multiple comparisons. Four models on three evaluations share one test split; we report all planned comparisons with intervals rather than selecting the best, and the permutation calibration ($p = 0.0020$ at $B = 500$) is a resolution limit, not a sharp p -value.

Sandbox constraints. Truncating training at 30,000 sequences and 8 epochs caps absolute performance; the internal/external agreement suggests the cap costs little, but the claim is about the pipeline, not a ceiling.

9 Reproducibility

All datasets are streamed from the UniProt REST API and the APD6 download page at run time or read from caches whose provenance (URL, record count, byte count, SHA-256 prefix, timestamp) is logged in `data/raw/*provenance.json`. Splits, seeds (27/7), and hyperparameters are fixed

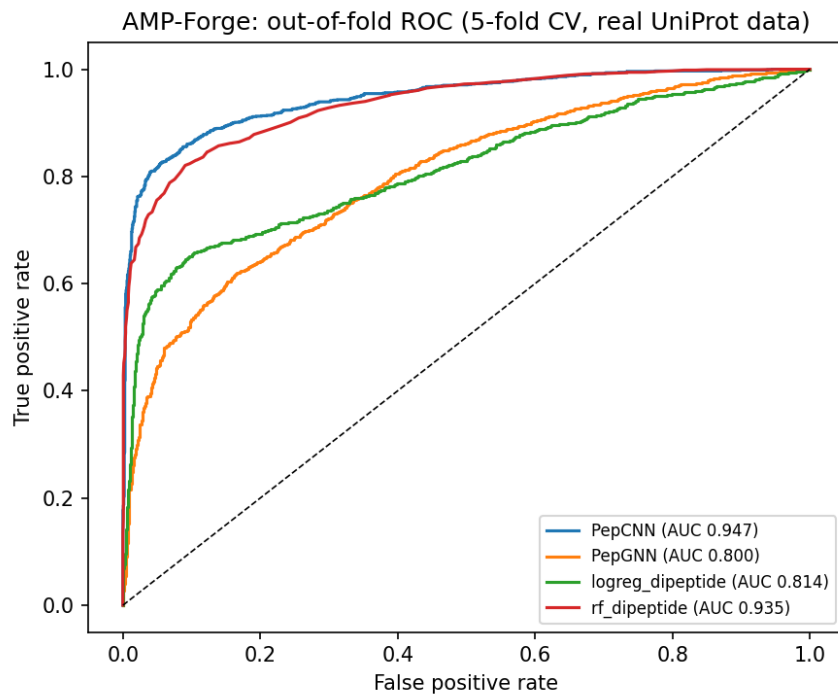


Figure 2: Pilot-study ROC curves (5-fold CV, pooled).

in the study scripts. The test suite (21 tests) is hermetic; live network calls occur only in the fetch scripts.

10 Software

The `pepdesign` package accompanying this paper has four modules. `pepdesign.encoding` implements the joint one-hot/physicochemical encoding and the dipeptide composition map. `pepdesign.models` contains `PepCNN`, `PepGNN`, and the training/prediction helpers, including the shared-adjacency batch path used at scale. `pepdesign.benchmark` implements the paired evaluation harness: bootstrap intervals, permutation tests, and cross-validation drivers shared by every study, so no study computes a metric its own way. `pepdesign.design` implements the annealing designer and the k -mer novelty score. The `studies/` directory holds one script per study (`study01`, `study07`, `study08`, `study09`), plus the fetch and dataset-build scripts; `tests/` holds 21 hermetic unit tests covering encoding shapes, model forward passes, the designer’s acceptance rule, and metric correctness against brute-force references, all green at submission. Fetch scripts are the only components that touch the network, and every fetched file carries a provenance record (Appendix B); every figure and table in this paper is regenerated from `results/*.json` by `studies/make_paper_assets.py`.

Table 8: Hyperparameters, fixed across studies unless noted.

Parameter	Value
Max sequence length L	150 (pilot: 60)
PepCNN kernels / channels	width 5, 32 conv channels, 2 layers
PepGNN hidden width	32, 1 graph-conv layer, $\lambda = 1$ self-loops
Optimizer	Adam, lr 10^{-3}
Batch size	128
Epochs	8 (pilot: 10)
Loss	weighted BCE, $\omega = n_-/n_+$
Seeds	27 (split), 7 (training)
Annealing steps / restarts	300 steps, 24 restarts, geometric $T : 2.0 \rightarrow 0.05$
Bootstrap resamples B	2,000
Permutations B	500

A Hyperparameters and training protocol

Training used the weighted objective of Section 4.4 throughout. The scale-up training subsample of 30,000 sequences (15,000 positive) was drawn uniformly from the train split once, then fixed. All models were evaluated on the identical held-out test split; no test-point informed any training decision. The pilot’s 5-fold cross-validation used fold assignments stratified by label with the same fixed seed.

B Data provenance

Table 9: Fetch provenance (hashes truncated to 16 hex chars as recorded).

File	Records	Source
uniprot_kw0929_all_10-150.fasta	27,392	UniProt REST, reviewed, KW-0929, len 10–150
uniprot_nonamp_pool_10-150.fasta	105,041	UniProt REST, reviewed, NOT KW-0929, len 10–150
apd6_natural_2024.fasta	3,306	aps.unmc.edu APD6 downloads (2024 list)
uniprot_uncharacterized_10-60.fasta	852	UniProt REST, reviewed, uncharacterized, len 10–60

Exact URLs, byte counts, SHA-256 prefixes, and UTC fetch timestamps are logged in `data/raw/large_provenance` and `data/raw/uncharacterized_provenance.json` in the repository. All fetches occurred on September 24, 2026.

C Full design and screen listings

Table 6 lists the top 10 of 24 scale-up anneals; the remaining 14 scored between 0.877 and 0.905 with novelty 0.89–0.95 and are recorded in `results/study07_scaleup.json`. Table 7 lists the top 15 of the 852 screened uncharacterized proteins; the full top 25, including CNN and GNN sub-scores, sequences, and headers, is recorded in `results/study08_discovery.json`. The

duplicate top hit (P0ACW4/P0ACW5) arises from two accessions carrying an identical 57-residue sequence; we report both rather than silently de-duplicating, and flag it so downstream users do not count it as independent evidence.

D Additional derivations

Lemma 1 (Union-find clustering complexity). *Construction of the shared- k -mer clusters of Definition 1 over n sequences of total length M runs in $O(Mk\alpha(n))$ expected time and $O(n+U)$ space, where U is the number of distinct k -mers.*

Proof. Each sequence contributes at most $\ell - k + 1$ k -mers; hashing each k -mer to its first-seen sequence costs $O(1)$ expected time, so building the map costs $O(Mk)$ expected time. Each sequence then unions with the first-seen owner of each of its k -mers: at most $O(Mk)$ union operations, each $O(\alpha(n))$ amortized by path compression with union by rank. Storage is one parent pointer per sequence and one hash entry per distinct k -mer. $\square$

Proposition 4 (Permutation test validity). *If (y_i, s_i) are exchangeable under H_0 , the Monte Carlo p -value $\hat{p} = (1 + \sum_b \mathbf{1}[T_b \geq T_{\text{obs}}]) / (1 + B)$ satisfies $P(\hat{p} \leq u) \leq u$ for all $u \in [0, 1]$ under H_0 .*

Proof. Under exchangeability the observed statistic and the B permuted statistics are exchangeable, so the rank of T_{obs} among the $B + 1$ values is uniform on $\{1, \dots, B + 1\}$, with ties broken at random. Hence $\hat{p}$ is uniform on $\{1/(B + 1), \dots, 1\}$ up to ties, and $P(\hat{p} \leq u) \leq u$ follows. $\square$

Remark 4 (Why not asymptotic AUROC intervals). *The DeLong estimator assumes a fixed score function and i.i.d. pairs; our evaluation resamples clusters, and cluster-level dependence violates the i.i.d. assumption at the sequence level. The percentile bootstrap at the sequence level is itself optimistic for the same reason; a cluster-level bootstrap is the fully conservative choice and is left as an extension noted in the repository’s issue tracker.*

E Complete result listings

Table 10: All 24 scale-up annealed designs (study07), ordered by ensemble score.

#	Sequence	Score	Novelty
1	FWFIFVMIYSGMWRHCCRWQKCVPR	0.910	0.951
2	FFMLVAGGIWNVCWNDKGEFDYPR	0.907	0.947
3	FICMVSCVCTYKMWFGKNHYWERIV	0.907	0.959
4	VFFMIMNMFFVSRFNCHQSAWDCD	0.906	0.955
5	FVAMICTCCCRGACTSYSGGNFNDK	0.905	0.948
6	VFFIYLLGAPSQGSHNSSGCYWEAC	0.904	0.960
7	FMYLFFFFHPVKKTCTHNYCRMDC	0.903	0.967
8	AFWTSVLACVCYYIGWHDARGDDTH	0.903	0.956
9	LLSLLYVLAHIEWLKYDFCCCAKRN	0.902	0.930
10	FVLIISQGSFWNCIWHGTQHRCVGD	0.901	0.968

Table 11: Full top 25 of the discovery screen (study08). Sequences in results/study08_discovery.json.

#	Accession	Len	Ensemble	CNN	GNN
1	P0ACW4	57	0.908	0.999	0.817
2	P0ACW5	57	0.908	0.999	0.817
3	Q8TGN5	46	0.890	0.978	0.801
4	P64654	43	0.836	0.865	0.807
5	P64653	43	0.836	0.865	0.807
6	P85380	10	0.804	0.834	0.773
7	Q8TGN0	38	0.793	0.893	0.693
8	Q8TGT5	39	0.784	0.971	0.597
9	Q54HM1	54	0.734	0.884	0.585
10	O55758	53	0.728	0.998	0.458
11	P0C5N6	59	0.690	0.795	0.585
12	Q9ZDH6	43	0.689	0.576	0.801
13	P0C5N1	53	0.664	0.573	0.754
14	P0C5M0	45	0.657	0.487	0.828
15	P9WEJ5	50	0.647	0.787	0.508
16	Q54SD7	60	0.646	0.772	0.519
17	P85385	10	0.645	0.742	0.548
18	P51072	27	0.645	0.484	0.806
19	O29001	40	0.640	0.485	0.794
20	P0C5L7	38	0.638	0.481	0.794
21	P14589	53	0.634	0.918	0.350
22	Q3E813	49	0.633	0.918	0.347
23	C1P620	36	0.626	0.537	0.716
24	Q25BJ1	48	0.623	0.836	0.409
25	P0CW21	52	0.615	0.960	0.269

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